Data Governance, Compliance and Ethics



WEEK 10: FAIRNESS, ACCOUNTABILITY, AND TRANSPARENCY OF ALGORITHMIC SYSTEMS (PART 1)

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Overview of course content

1. Data Governance and Management (Weeks 1-3)
2. Privacy and Data Protection (Weeks 4-6)
3. Ethical Issues Pertaining to Data (Weeks 7-9)*
4. Fairness, Accountability, and Transparency of Algorithmic Systems (Weeks 10-11)

Agenda

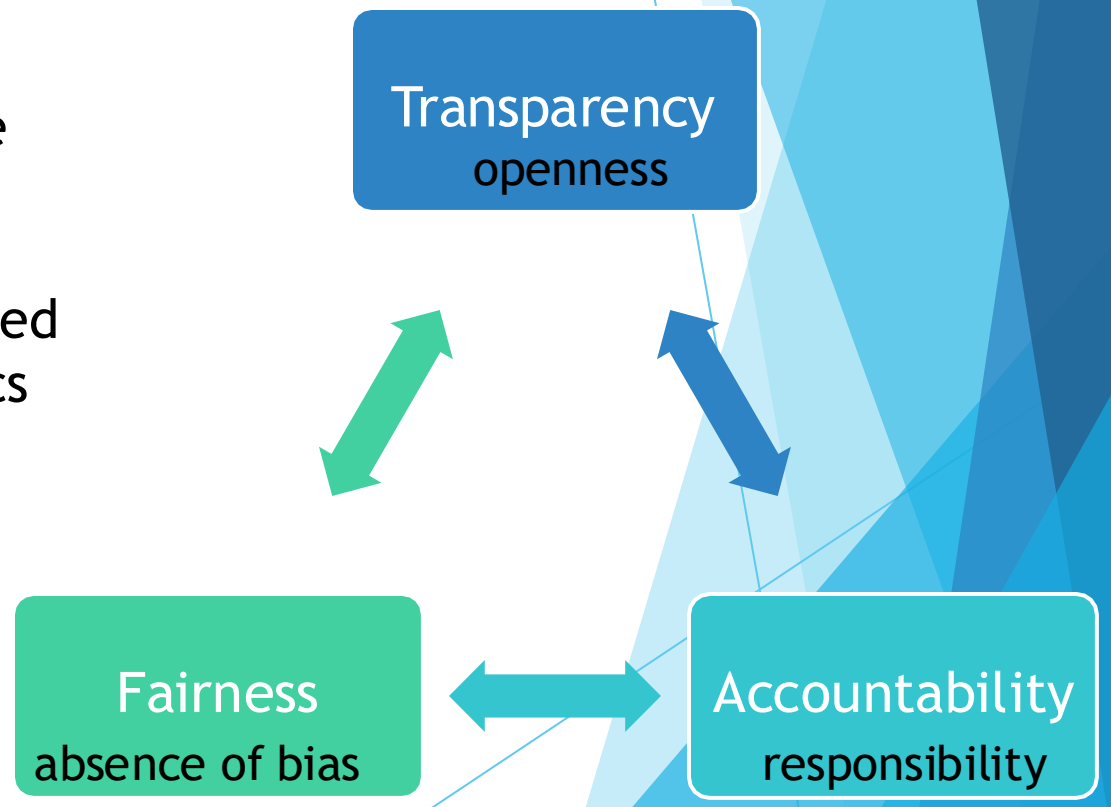
- The meaning of fairness with respect to algorithmic systems. Techniques and models for fairness-aware data mining, information retrieval, recommendation, etc. Legal, social, and philosophical models of fairness. Specification of mathematical objectives with respect to fairness. Perceptions of algorithmic bias and unfairness. Interventions to mitigate biases in systems, or discourage biased behaviour from users.
- The meaning of accountability with respect to algorithmic systems. Processes and strategies for developing accountable systems. Methods and tools and standards for ensuring that algorithms comply with fairness policies (e.g., IEEE P7003 TM).
- "The meaning of transparency with respect to algorithmic systems.. Explanations for algorithmic logic and outputs. Trade-offs between privacy and transparency. Tools and methodologies for conducting algorithm audits. Frameworks for conducting ethical and legal algorithm audits. Empirical results from algorithm audits."

Fairness, Accountability and Transparency of Algorithms

Fairness, Accountability and Transparency – way of examining critically one of the principal concerns about the use of algorithms in society (as part of the broader socio-technical system)

"In principle, **fairness** is the absence of any bias based on an individual's inherent or acquired characteristics that are irrelevant in the particular context of decision-making."

- To assess fairness, we need transparency
- To enforce fairness, we need accountability

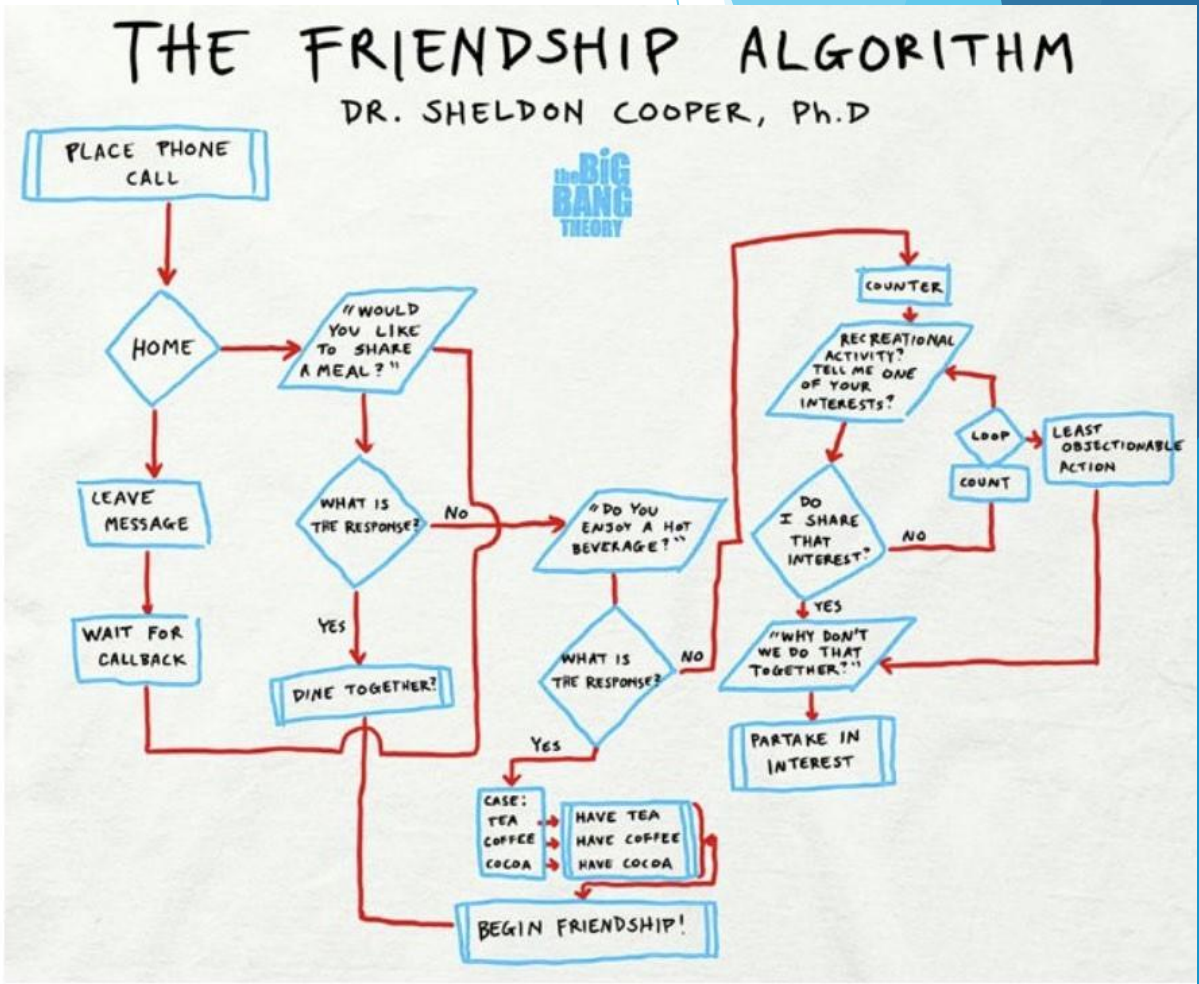


What Is an Algorithm?

“Algorithm” refers to a set of precise instructions or rules regarding actions to be taken in solving a predefined problem

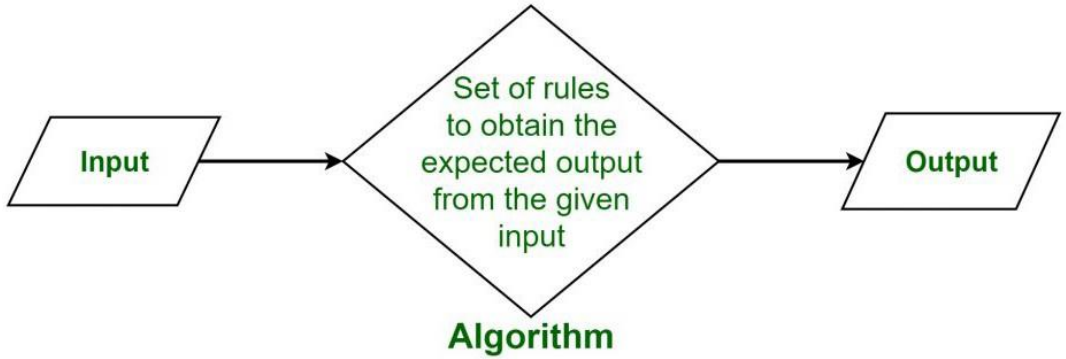


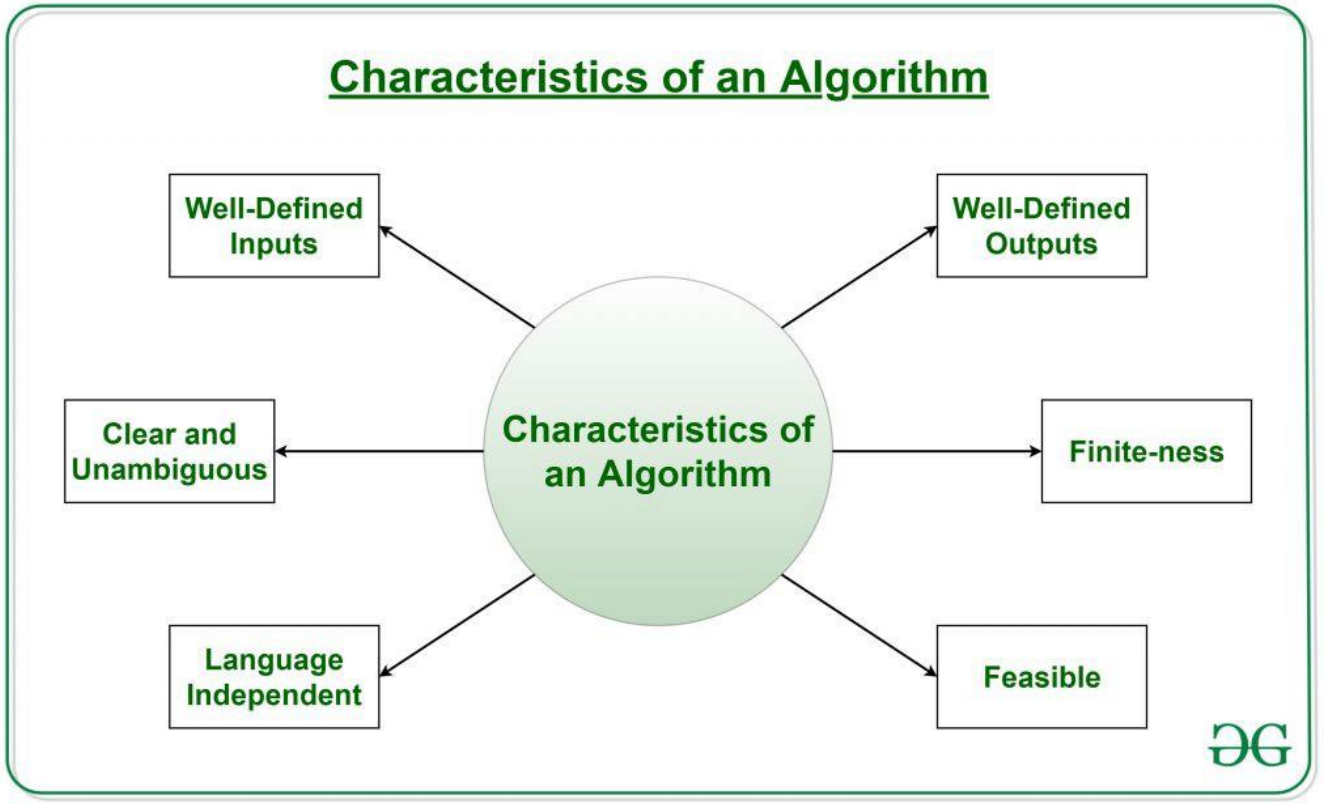
The big bang theory



An Algorithmic System

- **“Algorithm”** refers to a set of precise instructions or rules regarding actions to be taken in solving a predefined problem



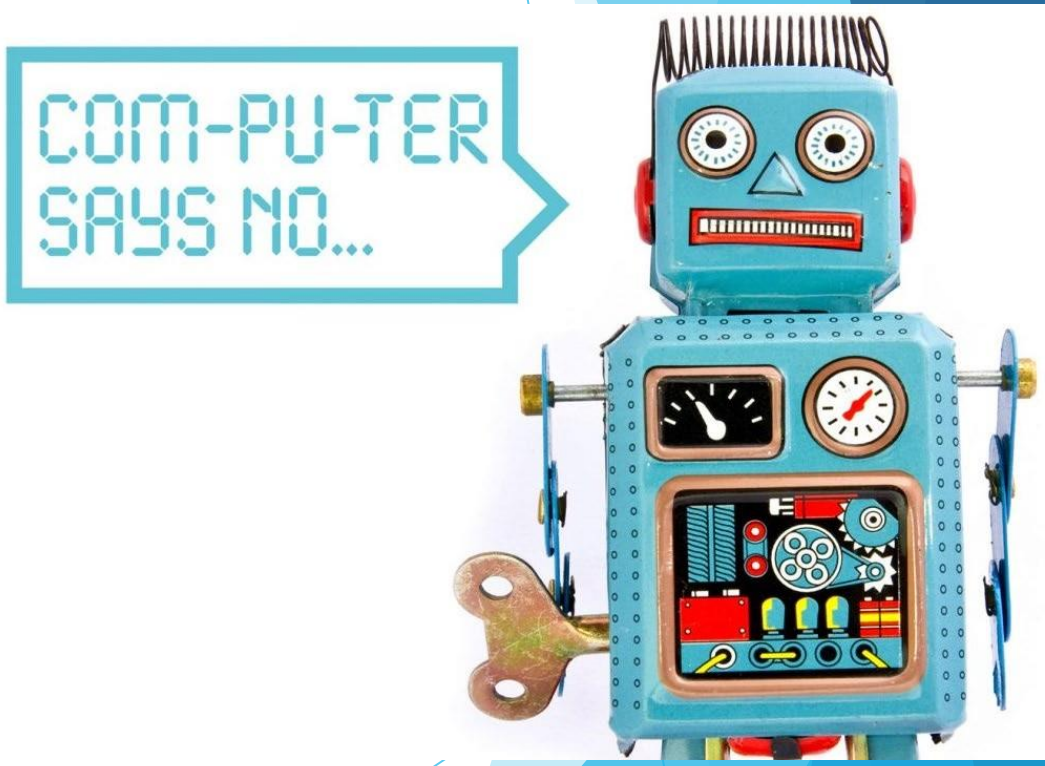


- **An algorithmic system** is **a system** comprised **of one or more algorithms** used in a software to collect and analyze data as well as draw conclusions **as part of a process** designed to solve a pre-defined problem

An Algorithmic System

- Algorithmic decision-making systems (AD-M sys) are ubiquitous across a wide variety of services. They rely on:
  - on complex learning methods
  - vast amounts of data
- **There is a growing concern that these automated decisions can lead to a lack of fairness**

- ML relies heavily on data. It is a common misconception that AI is absolutely objective. AI is objective only in the sense of learning what human teaches.

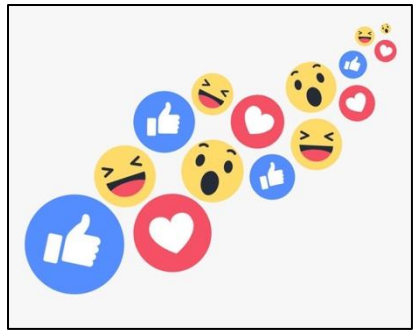


The Use of Algorithms

Algorithms can be used for:

- Mapping your online world

To predict what customers might want to buy/watch



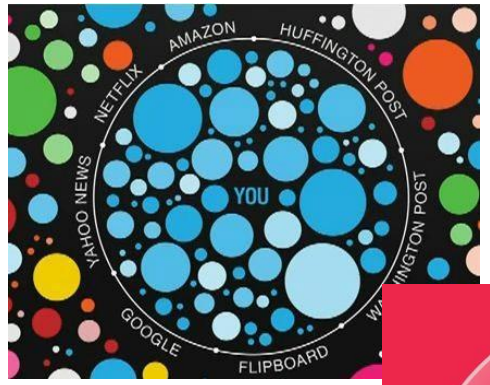
To conduct sentiment analysis of posts



To enable smart assistants

- Analyzing what is in your personal filter bubble

Not everyone gets the same results online





The Use of Algorithms

What happens when the output of decision-making algorithms have significant societal impact?

Loans



Issues have arisen regarding the fairness of these algorithmic decisions

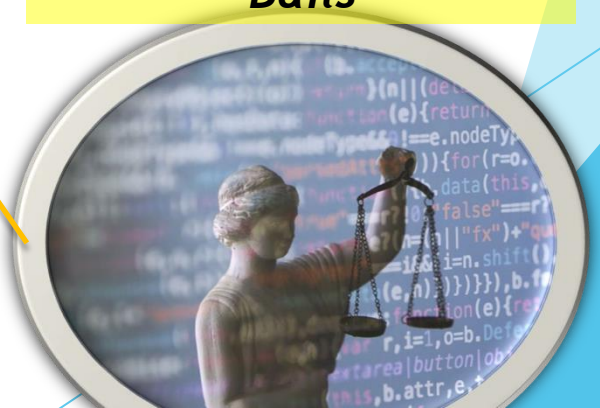
Hiring



University Admissions



Bails



Issues Arisen Regarding The Fairness Of Algorithms

Bails



- In 2016, ProPublica analyzed the algorithm used for recidivism predictions from COMPAS for criminal defendants
- The algorithm produces much higher false positive rate for black people than white people

	WHITE	AFRICAN AMERICAN
Labeled Higher Risk, But Didn't Re-Offend	23.5%	44.9%
Labeled Lower Risk, Yet Did Re-Offend	47.7%	28.0%

Overall, Northpointe's assessment tool correctly predicts recidivism 61 percent of the time. But blacks are almost twice as likely as whites to be labeled a higher risk but not actually re-offend. It makes the opposite mistake among whites: They are much more likely than blacks to be labeled lower risk but go on to commit other crimes. (Source: ProPublica analysis of data from Broward County, Fla.)

Fig2: The bias in COMPAS. (from Larson et al. ProPublica, 2016)

Issues Arisen Regarding The Fairness Of Algorithms

Hiring



- The XING platform was found to rank less qualified male candidates higher than more qualified female candidates

Search query	Work experience	Education experience	Profile views	Candidate	Xing ranking
Brand Strategist	146	57	12992	male	1
Brand Strategist	327	0	4715	female	2
Brand Strategist	502	74	6978	male	3
Brand Strategist	444	56	1504	female	4
Brand Strategist	139	25	63	male	5
Brand Strategist	110	65	3479	female	6
Brand Strategist	12	73	846	male	7
Brand Strategist	99	41	3019	male	8
Brand Strategist	42	51	1359	female	9
Brand Strategist	220	102	17186	female	10

TABLE II: Top k results on www.xing.com (Jan 2017) for the job serach query "Brand Strategist".

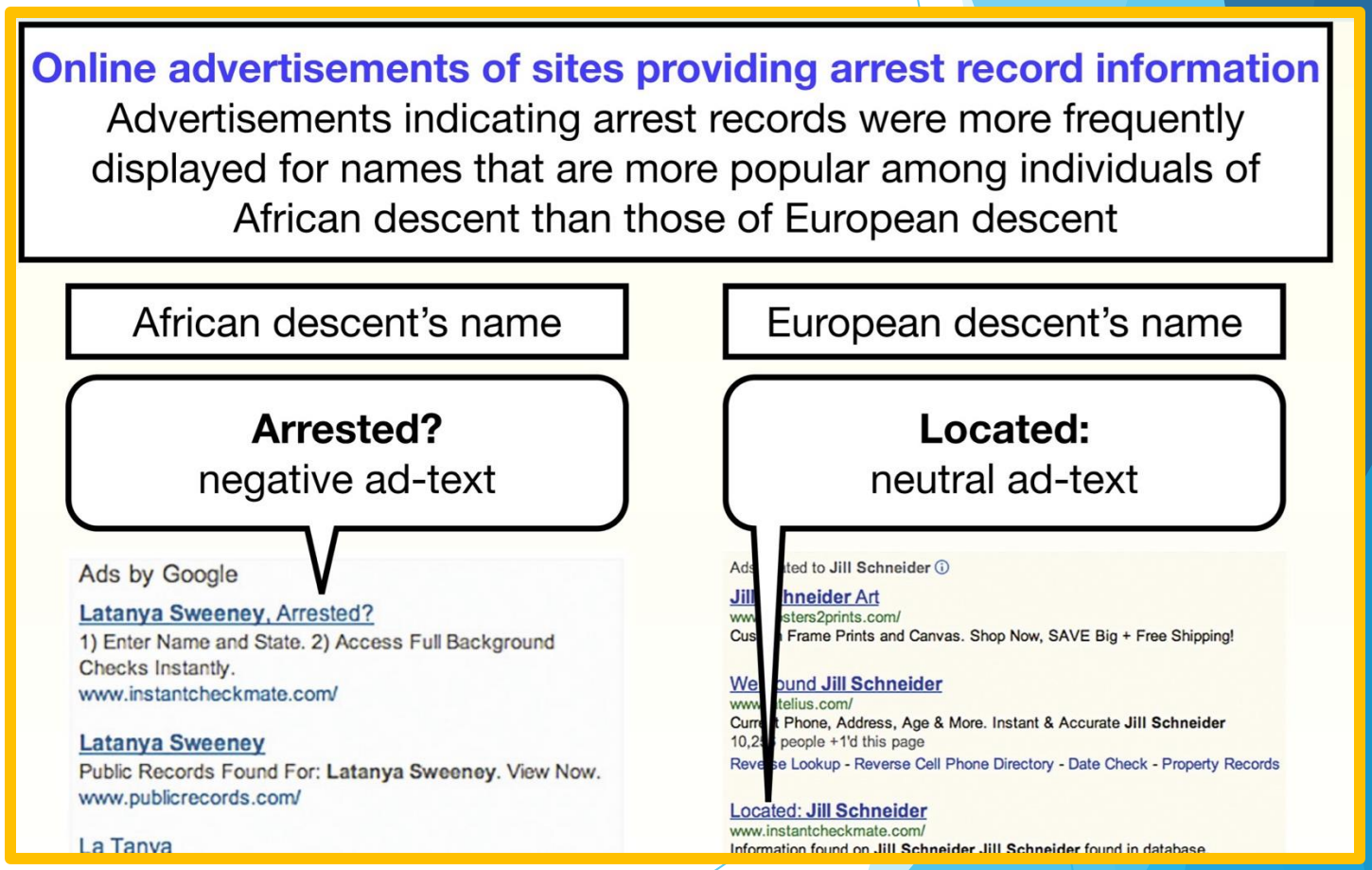
Fig3: The bias in the query for Brand Strategist from XING(from Lahoti et al. 2018).

Preethi Lahoti, Krishna P Gummadi, and GerhardWeikum. ifair: Learning individually fair data representations for algorithmic decision making. In 2019 IEEE 35th International Conference on Data Engineering (ICDE), pages 1334–1345. IEEE, 2019

Examples Of Algorithmic Biases

Bias in online ads

- Latanya Sweeney, Harvard researcher and former chief technology officer at the Federal Trade Commission (FTC), found that
  - **online search queries for African-American names were more likely to return ads to that person from a service that renders arrest records, as compared to the ad results for white names.**
  - the same differential treatment occurred in the micro-targeting of higher-interest credit cards and other financial products when the computer inferred that the subjects were African-Americans, despite having similar backgrounds to whites.



Sweeney, Latanya. "Discrimination in online ad delivery." Rochester, NY: Social Science Research Network, January 28, 2013. Available at https://papers.ssrn.com/abstract=2208240 (last accessed April 12, 2019).

Examples Of Algorithmic Biases

Bias in online recruitment tools

- Online retailer Amazon, whose global workforce is 60% male and where men hold 74% of the company's managerial positions, recently **discontinued use of a recruiting algorithm after discovering gender bias**
- The **data** that engineers used to create the algorithm were derived from the resumes submitted to Amazon over a 10-year period, which were **predominantly from white males**.
- **the AI software penalized any resume that contained the word "women's"** in the text and downgraded the resumes of women who attended women's colleges, resulting in gender bias



Vincent, James. "Amazon Reportedly Scraps Internal AI Recruiting Tool That Was Biased against Women." The Verge, October 10, 201 Available at https://www.theverge.com/2018/10/10/17958784/ai-recruiting-tool-bias-amazon-report (last accessed April 20, 2019).

Examples Of Algorithmic Biases

Bias in online recruitment tools

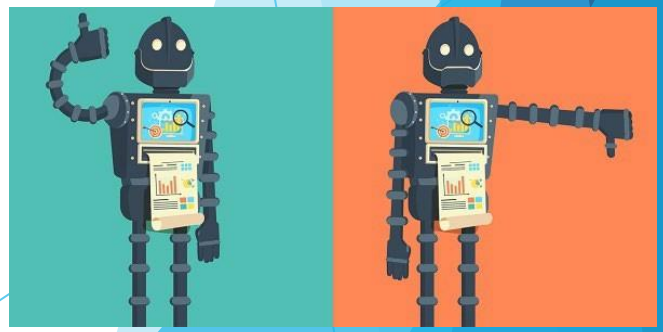
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Bias in word associations

- Princeton University researchers used off-the-shelf machine learning AI software to analyze and link 2.2 million words.
- They found that **European names were perceived as more pleasant** than those of African-Americans, and that **the words "woman" and "girl" were more likely to be associated with the arts instead of science and math**, which were most likely connected to males.[
- If the learned associations of these algorithms were used as part of a search-engine ranking algorithm or to generate word suggestions as part of an auto-complete tool, it could have a **cumulative effect of reinforcing racial and gender biases.**

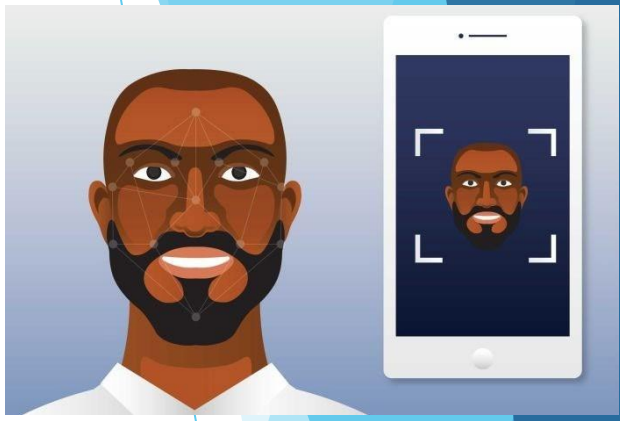


Hadhazy, Adam. "Biased Bots: Artificial-Intelligence Systems Echo Human Prejudices." Princeton University, April 18, 2017. Available at https://www.princeton.edu/news/2017/04/18/biased-bots-artificial-intelligence-systems-echo-human-prejudices (last accessed April 20, 2019).

Examples Of Algorithmic Biases

Bias in facial recognition technology

- the algorithms powering three commercially available facial recognition software systems were **failing to recognize darker-skinned complexions**
- most facial recognition training data sets are estimated to be more than 75% male and more than 80% white.
- When the person in the photo was a white man, the software was accurate 99% of the time at identifying the person as male. But error rate ranged from 20-34% for darker-skinned women female
- IBM and Microsoft committed to improving the accuracy of their recognition software for darker-skinned faces



Hardesty, Larry. "Study Finds Gender and Skin-Type Bias in Commercial Artificial-Intelligence Systems." MIT News, February 11, 2018. Available at http://news.mit.edu/2018/study-finds-gender-skin-type-bias-artificial-intelligence-systems-0212 (last accessed April 19, 2019).

Facial Recognition Software has consistently Shown Racial Bias When Identifying Faces





Google Photos mistakenly tagging two black friends in a selfie as gorillas

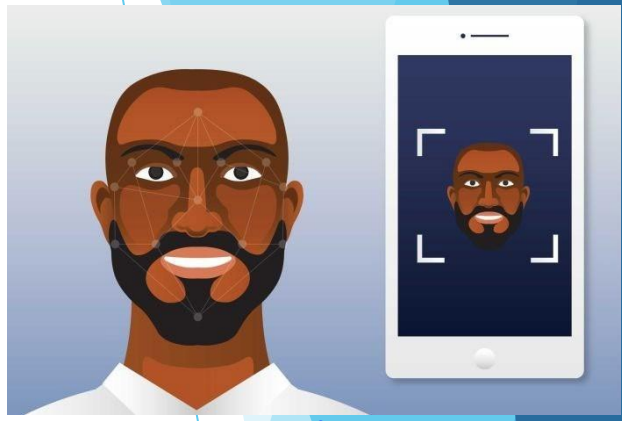


iPhone's face unlock not being able to differentiate between two colleagues in China

Examples Of Algorithmic Biases

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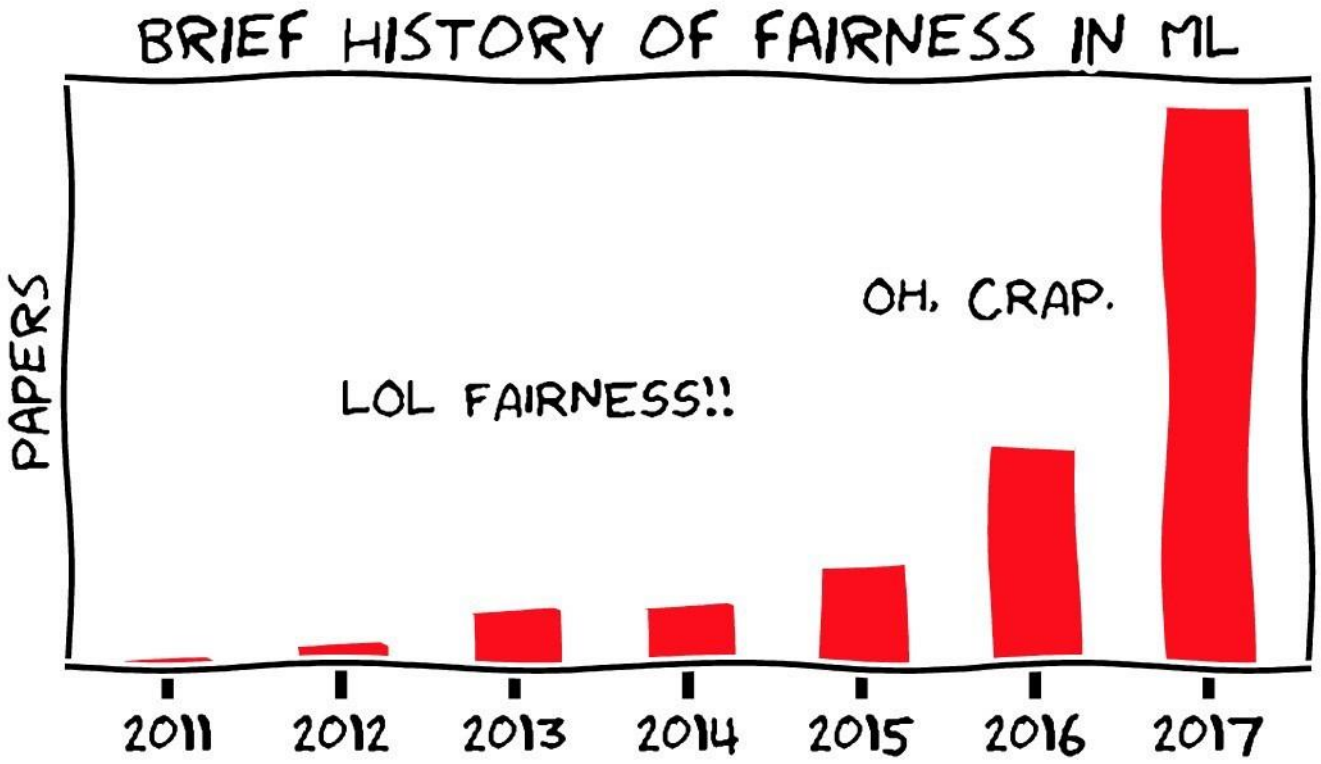
Bias in criminal justice algorithms

- COMPAS algorithm, which is used by judges to predict whether defendants should be detained or released on bail pending trial, was found to be biased against African-Americans
- The algorithm assigns a risk score to a defendant's likelihood to commit a future offense, relying on the voluminous data available on arrest records, defendant demographics, and other variables.
- Compared to whites who were equally likely to re-offend, African-Americans were more likely to be assigned a higher-risk score, resulting in longer periods of detention while awaiting trial



Corbett-Davies, Sam, Emma Pierson, Avi Feller, Sharad Goel, and Aziz Huq. "Algorithmic Decision Making and the Cost of Fairness." ArXiv:1701.08230 [Cs, Stat], January 27, 2017. https://doi.org/10.1145/3097983.309809

Fairness in Machine Learning Research



The number of publications on fairness from 2011 to 2017

ICML | 2022

Thirty-ninth International Conference on Machine Learning

Fairness, Accountability, and Transparency in Machine Learning

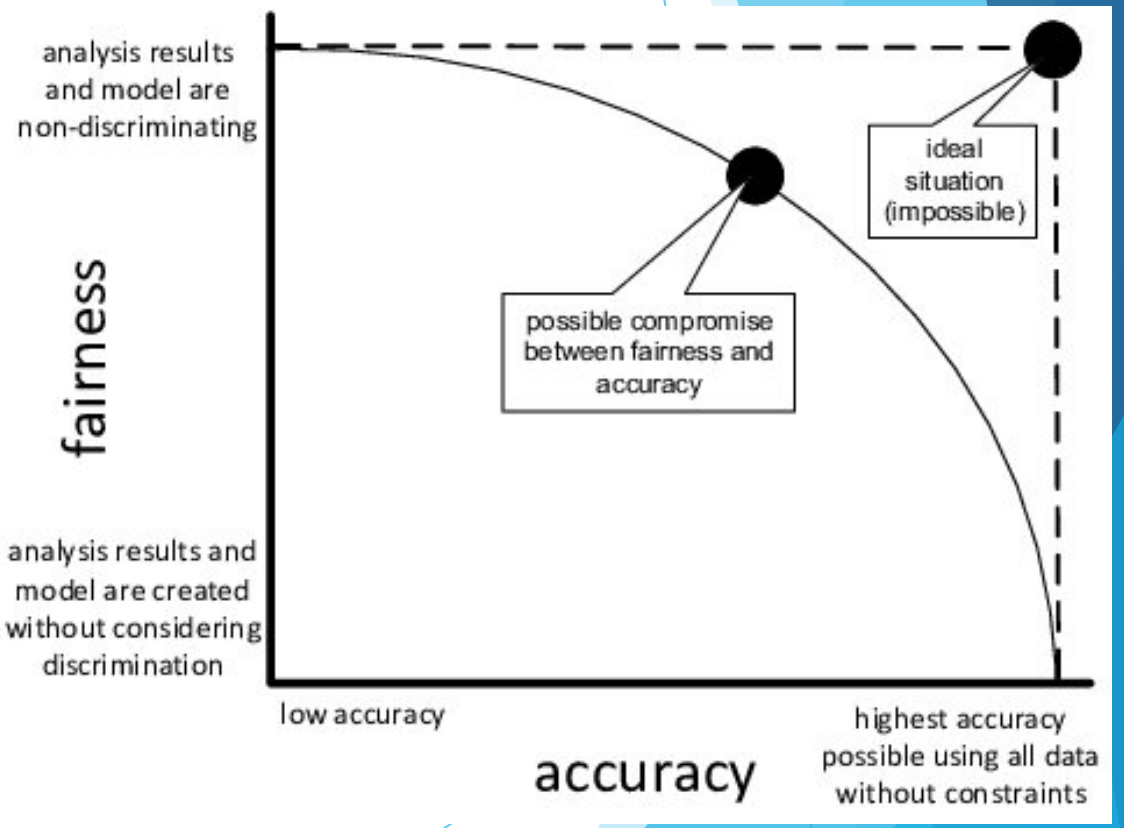
ACM Conference on Fairness, Accountability, and Transparency (ACM FAccT)

A computer science conference with a cross-disciplinary focus that brings together researchers and practitioners interested in fairness, accountability, and transparency in socio-technical systems.



The Meaning Of Fairness With Respect To Algorithmic Systems

- Studying *fairness in ML decision-making algorithms* means studying the algorithms **not only from a perspective of accuracy, but also from a perspective of fairness**
  - The difficulty revolves around **defining what fairness** means
  - In many cases these definitions have trade-offs with accuracy (i.e, achieving them means necessarily paying a price in terms of the model's accuracy)



Van Der Aalst, W.M., 2016. Green data science: using big data in an" environmentally friendly" manner. In *18th international conference on enterprise information systems (ICEIS 2016)* (pp. 9-21). SCITEPRESS-Science and Technology Publications, Lda..

Definition of Fairness

How People Perceive The Fairness?

In principle, fairness is the absence of any bias based on an individual's inherent or acquired characteristics that are irrelevant in the particular context of decision-making

#1 **Treating similar individuals similarly**: This is determined by a similarity distance metric (applied to certain attributes) which represents a notion of ground truth in regard to the decision context.

->*Individuals with similar repayment rates should receive similar amounts of money*



#2 **Never favor a worse individual over a better one**: This definition promotes meritocracy with respect to the candidate's inherent quality

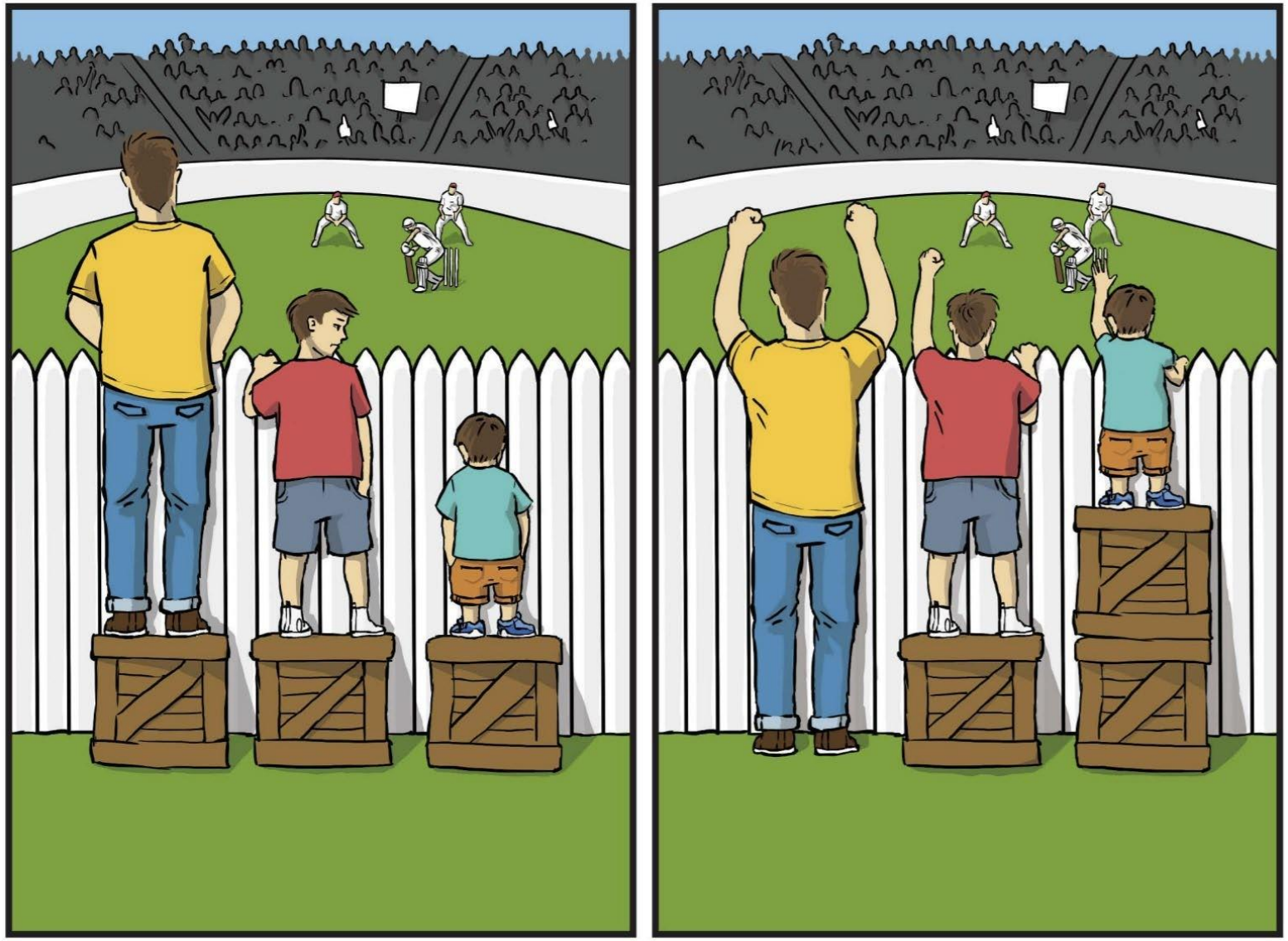
-> *An individual with a higher repayment rate should obtain at least as much money as her peer*

#3 **Calibrated fairness**: selects individuals in proportion to their merit

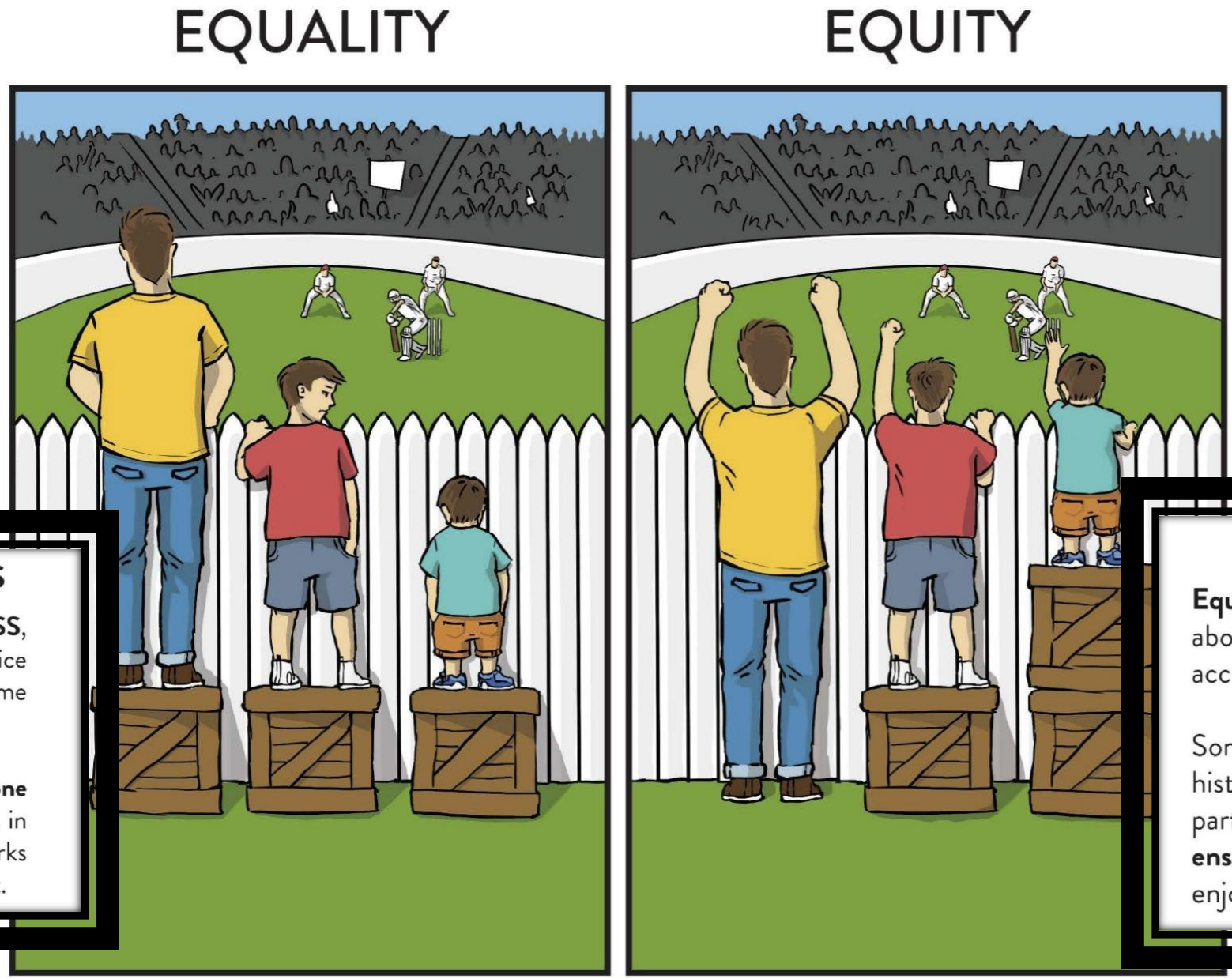
-> *two individuals with repayment rates r1 and r2, respectively, should obtain r1/(r1 + r2) and r2/(r1 + r2) amount of money, respectively*

Saxena, N.A., Huang, K., DeFilippis, E., Radanovic, G., Parkes, D.C. and Liu, Y., 2019, January. How do fairness definitions fare? Examining public attitudes towards algorithmic definitions of fairness. In *Proceedings of the 2019 AAAI/ACM Conference on AI, Ethics, and Society* (pp. 99-106).

What Is Fairness?



What Is Fairness?



Equality = SAMENESS

Equality is about SAMENESS, it promotes fairness and justice by giving everyone the same thing.

BUT it can **only work IF everone starts from the SAME place**, in this example equality only works if everyone is the same height.

Equity = FAIRNESS

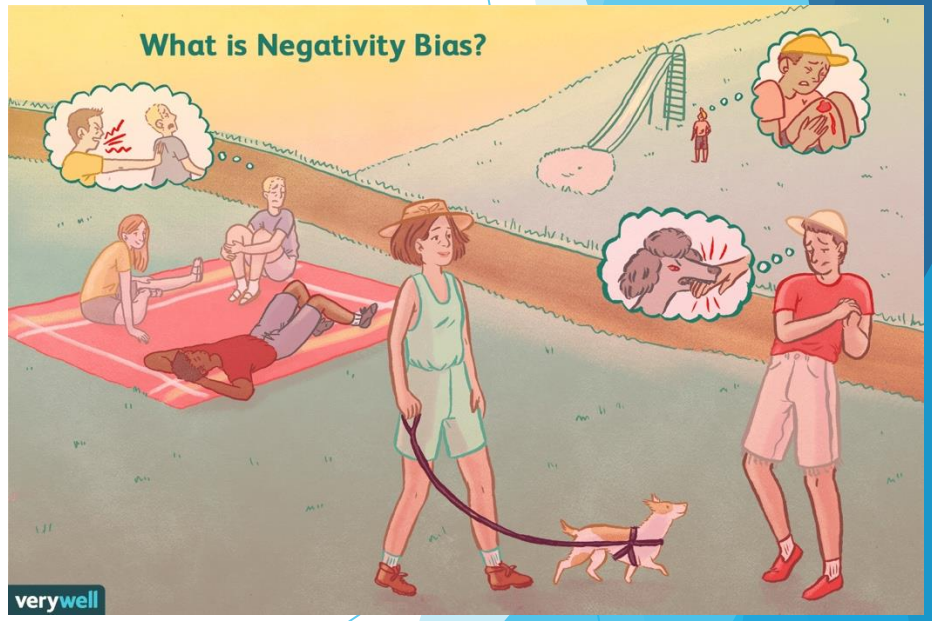
Equity is about FAIRNESS, it's about making sure people get access to the same opportunities.

Sometimes our differences and/or history can make barriers to participation, so we must **FIRST ensure EQUITY** before we can enjoy equality.

Historical Human Biases

- Historical human biases are shaped by pervasive and often deeply embedded prejudices against certain groups, which can lead to their reproduction and amplification in computer models.
- As humans, we tend to:
  a) Remember traumatic experiences better than positive ones
  b) Recall insults better than praise.
  c) React more strongly to negative stimuli.
  d) Think about negative things more frequently than positive ones.
  e) Respond more strongly to negative events than to equally positive ones.

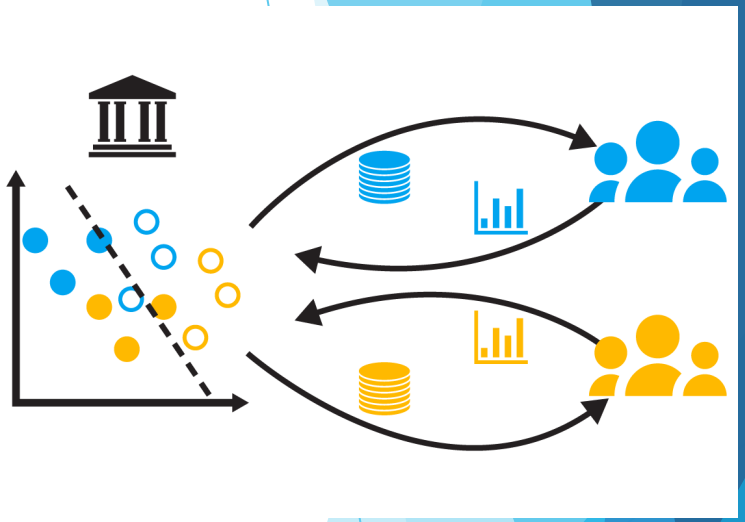
  - these realities will be reflected in the training data
  - If historical biases are factored into the model, it will make the same kinds of wrong judgments that people do.



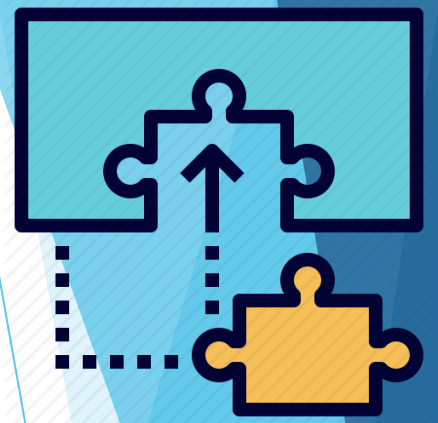
Negativity Bias: Negative events have a greater impact on our brains than positive ones.

Results of Historical Human Biases

- Further, **human biases can be reinforced and perpetuated** without the user's knowledge.
  - For example, clicking on ads without realizing that people will continue to receive such predatory online suggestions
- In this and other cases, the algorithm may never accumulate counter-factual ad suggestions.
- Thus, **it is important for algorithm designers and operators to watch for such potential negative feedback** loops that cause an algorithm to become increasingly biased over time.



Potential Causes Of Unfairness



1. Biases **already included in the datasets**, which are based on:
   - biased device measurements
   - historically biased human decisions - unfair labelling by annotators
   - erroneous reports or other reasons
2. Biases caused by **missing data**
3. Sample selection bias - datasets are not representative of the target population
4. **Limited features** - features may be less informative or reliably collected for minority group
5. Size disparity – **unbalanced dataset**
6. Biases caused by **“proxy" attributes** (non-sensitive attributes that can be exploited to derive sensitive attribute)

Pessach, D. and Shmueli, E., 2020. Algorithmic fairness. *arXiv preprint arXiv:2001.09784*.

Example of problems in datasets

siliconrepublic TECHNOLOGY SCIENCE CAREERS MORE LOGIN SUBSCRIBE

80m images used to train AI pulled after researchers find string of racist terms

by Colm Gorey

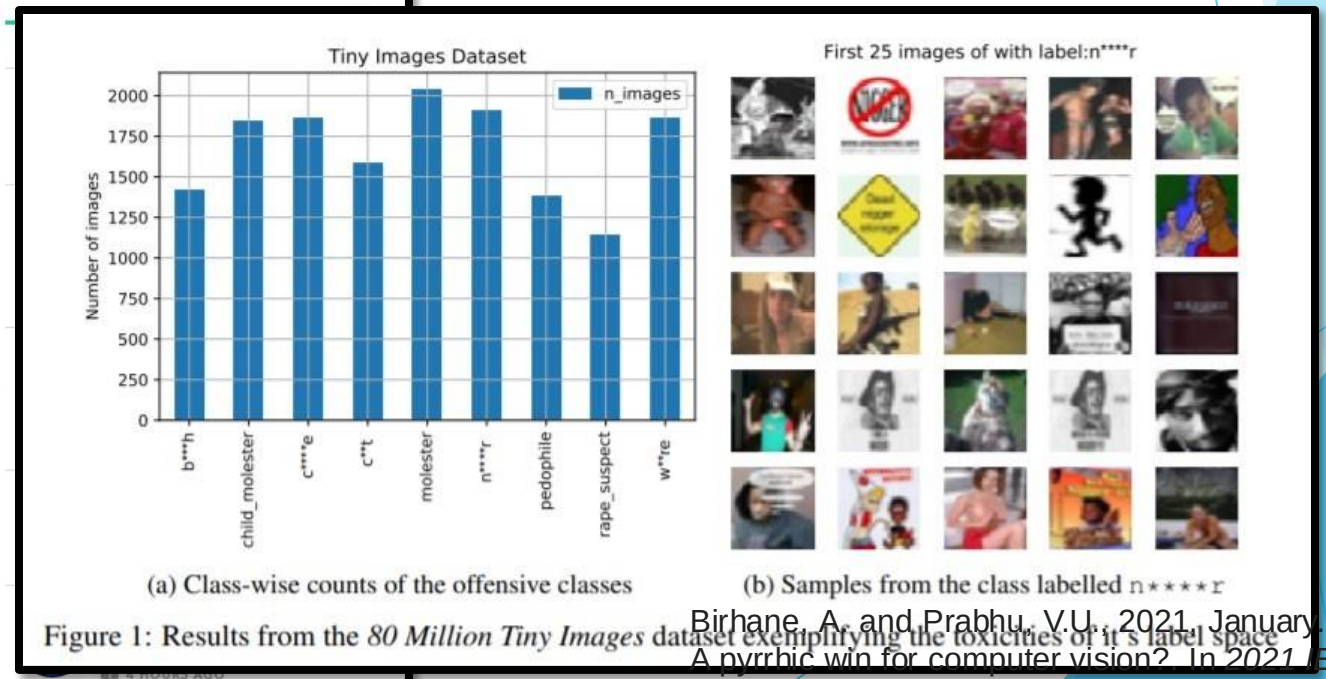
13 JUL 2020 SAVE ARTICLE



Lero and UCD PhD candidate Abeba Birhane. Image: Abeba Birhane

Lero and UCD researcher Abeba Birhane has helped uncover racist and misogynistic terms in an MIT image library that was used to train AI.

‘80 Million Tiny Images’ dataset may have contaminated AI systems with racist, misogynistic and other slurs.



(a) Class-wise counts of the offensive classes (b) Samples from the class labelled n****r

Figure 1: Results from the *80 Million Tiny Images* dataset exemplifying the toxicities of it's label space

Birhane, A. and Prabhu, V.U., 2021, January. Large image datasets: A pyrrhic win for computer vision?. In *2021 IEEE Winter Conference on Applications of Computer Vision (WACV)* (pp. 1536-1546). IEEE.

Example of problems in datasets

siliconrepublic

80m image
after resea
racist term

by Colm Gorey

13 JUL 2020 SAVE ARTICLE



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Lero and UCD researcher Abeba Birhane has helped uncover racist and misogynistic terms in an MIT image library that was used to train AI.

June 29th, 2020

It has been brought to our attention [1] that the Tiny Images dataset contains some derogatory terms as categories and offensive images. This was a consequence of the automated data collection procedure that relied on nouns from WordNet. We are greatly concerned by this and apologize to those who may have been affected.

The dataset is too large (80 million images) and the images are so small (32 x 32 pixels) that it can be difficult for people to visually recognize its content. Therefore, manual inspection, even if feasible, will not guarantee that offensive images can be completely removed.

We therefore have decided to formally withdraw the dataset. It has been taken offline and it will not be put back online. We ask the community to refrain from using it in future and also delete any existing copies of the dataset that may have been downloaded.

How it was constructed: The dataset was created in 2006 and contains 53,464 different nouns, directly copied from Wordnet. Those terms were then used to automatically download images of the corresponding noun from Internet search engines at the time (using the available filters at the time) to collect the 80 million images (at tiny 32x32 resolution; the original high-res versions were never stored).

Why it is important to withdraw the dataset: biases, offensive and prejudicial images, and derogatory terminology alienates an important part of our community -- precisely those that we are making efforts to include. It also contributes to harmful biases in AI systems trained on such data. Additionally, the presence of such prejudicial images hurts efforts to foster a culture of inclusivity in the computer vision community. This is extremely unfortunate and runs counter to the values that we strive to uphold.

Yours Sincerely,

Antonio Torralba, Rob Fergus, Bill Freeman.

taset
ystems
other



Figure 1: Results from the *80 Million Tiny Images* dataset exemplifying the toxicities of it's label space

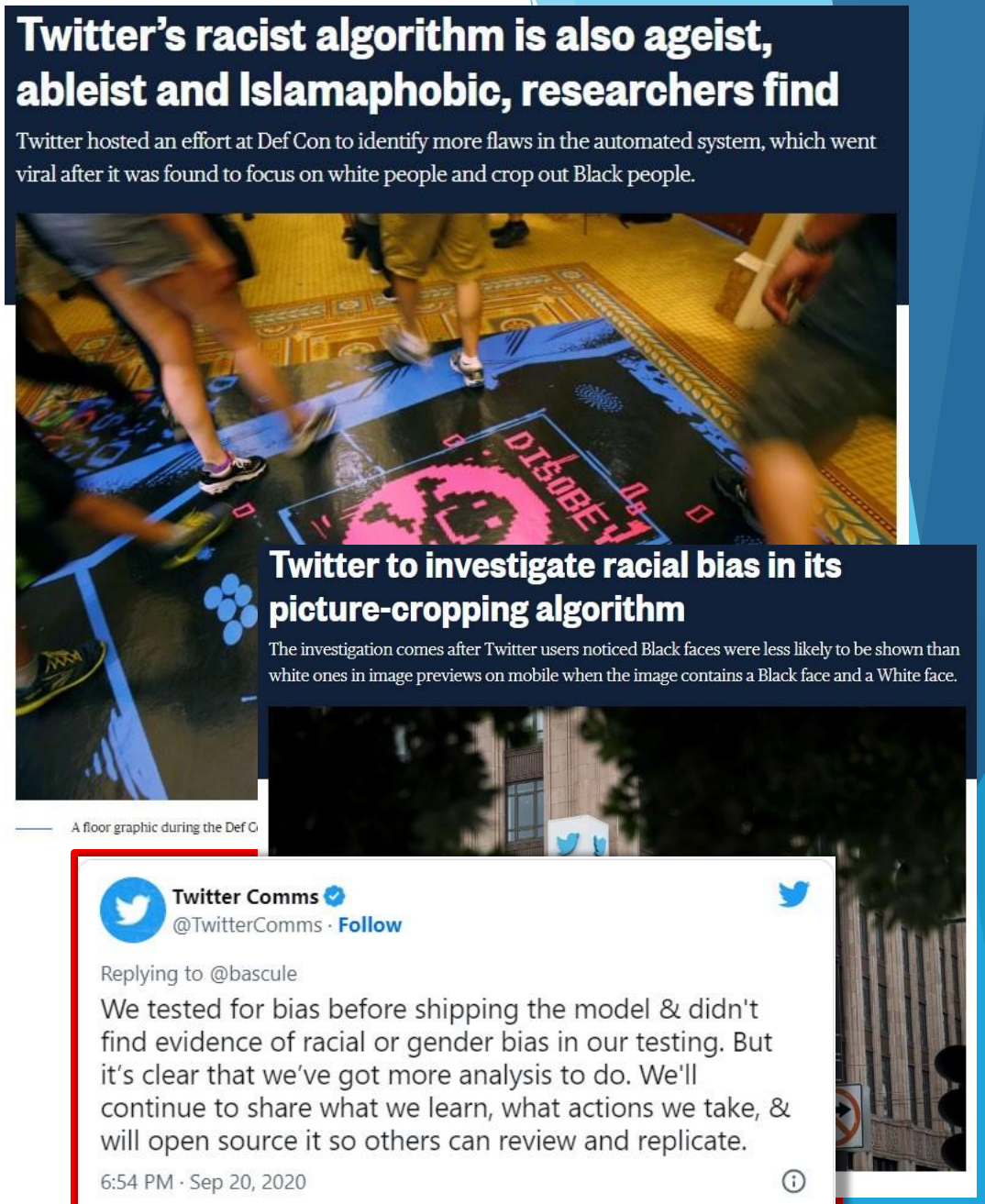
Birhane, A., and Prabhu, V. U. (2021, January). Large image datasets: A pyrrhic win for computer vision?. In *2021 IEEE Winter Conference on Applications of Computer Vision (WACV)* (pp. 1536-1546). IEEE.

Prejudices in AI systems

A Twitter image-cropping algorithm that went viral last year when users discovered it preferred white people to Black people was also coded **with implicit bias against a number of** other **groups**

Researchers looked at how the algorithm, which automatically edited photos to focus on people's faces, dealt with a variety of different people and saw evidence that Muslims, people with disabilities and older people faced similar discrimination.
The same artificial intelligence had learned to ignore people with white or gray hair, who wore headscarves for religious or other reasons, or who used wheelchairs, researchers said.

When companies train artificial intelligence to learn from their users' behavior — like seeing what kind of photos the majority of users are more likely to click on **— the AI systems can internalize prejudices that they would never intentionally write into a program.**



https://www.nbcnews.com/tech/tech-news/twitters-racist-algorithm-also-ageist-ableist-islamaphobic-researchers-rcna1632

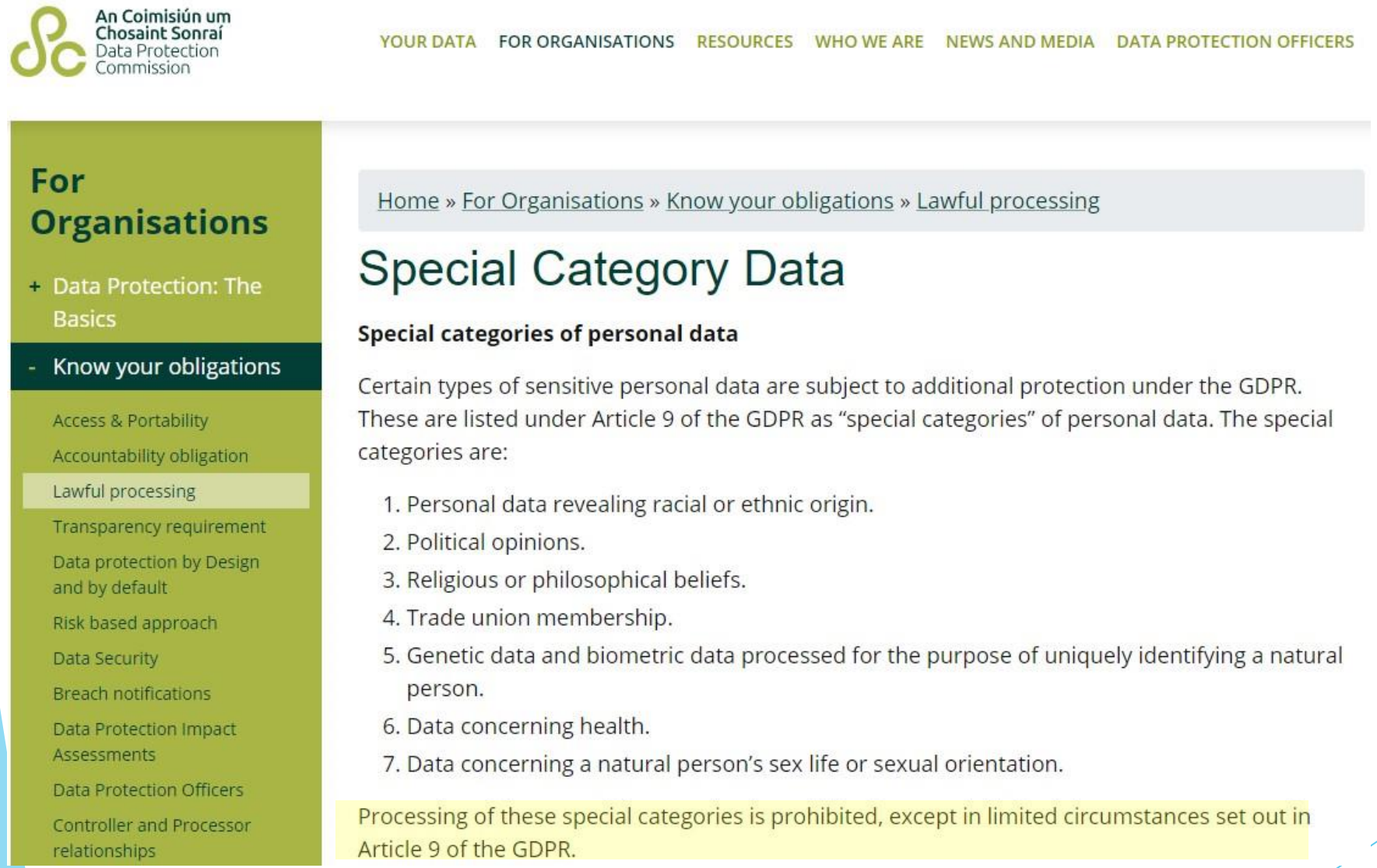
Detecting And Mitigating Bias (1/4)

- Bias detection should begin with **careful handling of the sensitive information of users**, including data that identify a person's membership in a federally protected group (e.g., race, gender).
- In some cases, operators of algorithms may also worry about a person's membership in some other group if they are also susceptible to unfair outcomes.

  An examples of this could be college admission officers worrying about the algorithm's exclusion of applicants from lower-income or rural areas; these are individuals who may be not federally protected but do have susceptibility to certain harms (e.g., financial hardships).
- Recent research has proposed to **use encryption over sensitive attributes** to improve fairness while still allowing the model to be trained, checked, or have its outputs verified and held to account



Special Category Data



Examples:

- Explicit consent
- necessary for the purposes of carrying out the obligations and exercising specific rights of the controller or of the data subject in the field of employment and social security
- vital interest of the data subject
- legitimate activities
- personal data which are manifestly made public by the data subject, etc.

https://gdpr-info.eu/art-9-gdpr/

https://www.dataprotection.ie/en/organisations/know-your-obligations/lawful-processing/special-category-data

GDPR And FaccT

- **From GDPR Recital 71 (Profiling):** "In order to ensure fair and transparent processing in respect of the data subject, taking into account the specific circumstances and context in which the personal data are processed, **the controller should use appropriate mathematical or statistical procedures for the profiling**, implement technical and organisational measures appropriate to ensure, in particular, that factors which result in inaccuracies in personal data are corrected and the risk of errors is minimised, secure personal data in a manner that takes account of the potential risks involved for the interests and rights of the data subject and **that prevents, inter alia, discriminatory effects on natural persons on the basis of racial or ethnic origin, political opinion, religion or beliefs, trade union membership, genetic or health status or sexual orientation**, or that result in measures having such an effect."



GDPR And FaccT

- Recall that a data subject has a right to contest an individual algorithmic decision (GDPR Art.22), to receive notice of solely automated decision-making (GDPR Art.13), and to request access to "meaningful information about the logic involved" (GDPR Art.15). In this way, data controllers can be challenged and thus accountable for algorithmic decisions.
- GDPR Article 35 (Data protection impact assessment) further provides that where data processing (in particular using new technologies) is likely to result in a high risk to the rights and freedoms of natural persons, the controller shall, prior to the processing, carry out an assessment of the impact of the envisaged processing operations on the protection of personal data. A single assessment may address a set of similar processing operations that present similar high risks and that the controller consult with the data protection officer when carrying out a data protection impact assessment.
- It is argued (Kaminski and Malgieri, 2019) that this requirement together with the recital, provides a case that "**GDPR's version of an Algorithmic Impact Assessment (that is DPIA)** serves as a central connection between its two approaches to regulating algorithms: individual rights and systemic governance."

Kaminski, M. E. and Malgieri, G. (2019) 'Algorithmic Impact Assessments under the GDPR: Producing Multi-layered Explanations', *SSRN Electronic Journal*, pp. 1–29

Detecting And Mitigating Bias (2/4)

- Computer programmers normally examine the set of outputs that the algorithm produces to **check for anomalous results**.
  - Comparing outcomes for different groups can be a useful first step.
- This could even be done through simulations
  - companies consider the simulation of predictions (both true and false) before applying them to real-life scenarios



For example, if a job-matching algorithm's average score for male applicants is higher than that for women, further investigation and simulations could be warranted.

Detecting And Mitigating Bias (3/4)

- **Fairness-Aware Data Collection and Curation**
  - improved attention to identifying blind spots, biases and limitations in machine learning team and other actors (such as those responsible for outcome decisions /labelling)
- More proactive and holistic auditing processes
  - ✓ Code audits
  - ✓ Scraping audits
  - ✓ Sock puppet audits
  - ✓ Crowdsourced/collaborative audits
- Addressing issues detected during machine learning: systems cannot substitute for the decision-making that lies within the discretion of the human arbiter

Holstein, K. *et al.* (2019) 'Improving fairness in machine learning systems: What do industry practitioners need?', *Conference on Human Factors in Computing Systems - Proceedings*, pp. 1–16.

Detecting And Mitigating Bias (4/4)

- Ethical Frameworks
  - **Ethics Guidelines for Trustworthy AI**: (1) human agency and oversight, (2) technical robustness and safety, (3) privacy and data governance, (4) transparency, (5) diversity, nondiscrimination and fairness, (6) environmental and societal well-being, and (7) accountability
  - it is unethical to “unfairly discriminate”
- Algorithmic Impact Assessments (AIAs): evaluate the potential detrimental effects of an algorithm

Fairness Tools

- AI fairness global library portal site
- Fairlearn
- AI Fairness 360 by IBM (https://aif360.mybluemix.net/)
- audit-AI
- ML-fairness-gym by Google
- Algorithmic Fairness (black-box auditing / fairness-comparison) by the Algorithmic Fairness group
- delax
- TensorFlow fairness-indicators
- FAT-forensics
- fairml Sample implementations of FADM algorithms by Žliobaitė's group
- Conditional non discrimination by Žliobaitė
- DCUBE: Discrimination Discovery in Databases by Pedrescchi, Ruggieri, and Turini
- Learning Fair Classifiers by Zafar

https://www.kamishima.net/faml/

Agenda

- The meaning of fairness with respect to algorithmic systems. Techniques and models for fairness-aware data mining, information retrieval, recommendation, etc. Legal, social, and philosophical models of fairness. Specification of mathematical objectives with respect to fairness. Perceptions of algorithmic bias and unfairness. Interventions to mitigate biases in systems, or discourage biased behaviour from users.
- The meaning of accountability with respect to algorithmic systems. Processes and strategies for developing accountable systems. Methods and tools and standards for ensuring that algorithms comply with fairness policies (e.g., IEEE P7003 TM).
- "The meaning of transparency with respect to algorithmic systems.. Explanations for algorithmic logic and outputs. Trade-offs between privacy and transparency. Tools and methodologies for conducting algorithm audits. Frameworks for conducting ethical and legal algorithm audits. Empirical results from algorithm audits."